

Mahmoud Ramzy

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Summary

Computer and Systems Engineering graduate with hands-on experience across software development, AI, IT infrastructure, and systems. Strong foundation in OOP, Data Structures and Algorithms, Databases, Operating Systems, and Networking, with practical experience building full-stack applications and internal technology systems.

Education

Alexandria University

8/2019 - 9/2024

BSc. Computer and Systems Engineering

Last term GPA: 3.8

CGPA: 3.35

Graduation Project: A+

Skills

Languages: Java, Python, JavaScript, C/C++

Frontend: HTML, CSS, React.js, Tailwind CSS

Backend: Node.js, Spring Boot, FastAPI, REST APIs

Databases: MySQL, PostgreSQL, SQL

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, YOLO

Concepts: Data Structures, OOP, Design Patterns, Operating Systems, Database Design, Networking

Tools/Version Control: Git, GitHub, Docker, Linux

Experience

IT / Systems – TFC (Video Editing Company)

7/2026 – 9/2026

- Managed internal IT operations including hardware maintenance, NAS, shared folders, access permissions, and endpoint controls.
- Customized and extended the Plane project-management platform with company branding, voice comments, desktop notifications, and hourly time tracking.
- Designed and built TFC Monitor, a centralized Windows monitoring system using FastAPI, PostgreSQL, Grafana, Docker, and Windows services.

Programming Instructor – iSchool

5/2026 – 7/2026

- Taught Python, Web Development, UI/UX, and Scratch through practical lessons and exercises.
- Guided students in programming fundamentals, problem-solving, debugging, and project development.

Teaching Assistant – Alexandria University & Alexandria National University

9/2024 – 1/2025

- Taught and simplified core CS concepts (Discrete Math, Network Security, OOP) for undergraduate students, including assignment preparation and performance evaluation.

Projects

TFC Monitor (Code Link)

- Built a centralized Windows endpoint monitoring system for device health, processes, TCP connections, and application network traffic.
- Designed the architecture connecting Windows clients, FastAPI backend, PostgreSQL database, and Grafana dashboards.

Full Stack Restaurant Application (Code Link)

- Built a full-stack web application using React.js, Node.js, and MySQL.
- Developed responsive UI and integrated backend APIs while collaborating in a team of six using GitHub.

Weather Stations Monitoring (Code Link)

- Designed and developed a distributed IoT data pipeline for real-time weather monitoring and analytics.
- Implemented stream processing, custom BitCask storage, Parquet archival, real-time dashboards, and containerized deployment using Java, Kafka, Elasticsearch, Docker, and Kubernetes.

Graduation Project: Hate Speech Detection in Arabic Tweets (Report Link)

- Developed an NLP model for Arabic hate speech detection using preprocessing, data augmentation, and ensemble models.
- Improved F1-score from 0.528 to 0.5479 and achieved an A+ graduation project grade.

Movie Application (Live Demo)

- Built a responsive movie application using React, Vite, TMDB API, and Appwrite.
- Implemented a custom debounce hook to optimize search requests and improve API usage.

Certificates

Deep Learning Specialization – Coursera 8/2023

- Finished 5 courses of the specialization on Coursera, covering neural networks, CNNs, RNNs, and hyperparameter tuning. **Verification Link**

Professional Front-End Web Development – Udacity 8/2022

- Completed practical training in APIs, semantic HTML, responsive CSS, and advanced JavaScript. **Verification Link**

Military Status

Completed – Egyptian Navy 3/2026